COURSE: BUSINESS INTELLIGENCE (IT730A)

BUSINESS INTELLIGENCE & ANALYTICS

PRESENTED BY
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Agenda

BI & Analytics

1. What is that?
2. Why is it needed?
3. Application Areas
4. Evolution

Business Intelligence

Techniques, technologies, systems, practices, methodologies & applications that analyze critical business data to help and enterprise **better understand its business and market and make timely business decisions**.

The discovery of meaningful patterns –new and novel information and knowledge– in data.

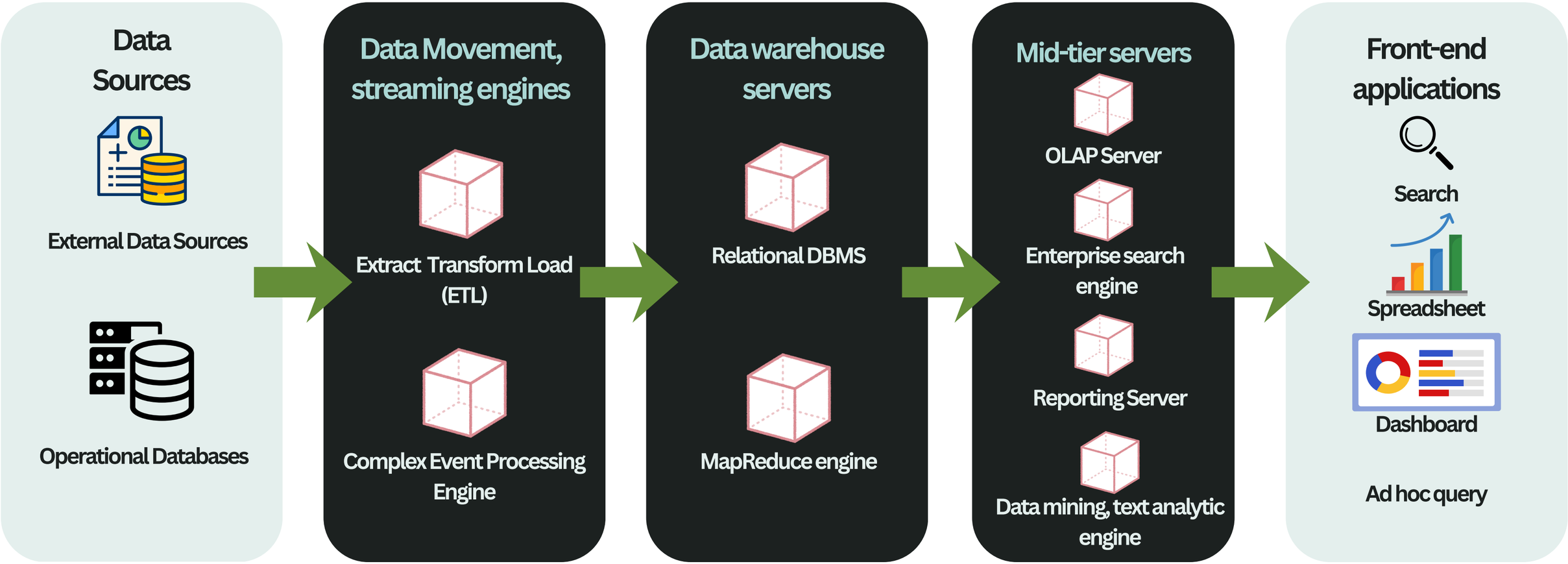
The Blurry Definition of Analytics

The analytics term is used imprecisely.

- Sometimes, it is employed interchangeably with **business intelligence**.
- Another interpretation is that if you view BI as “getting data in” (to the warehouse) and “getting data out” (data access and analysis), analytics is the **analysis part of BI**.
- Or finally, analytics is the **“rocket science” algorithms** and methods used to find patterns in data.

It is useful to think of descriptive, predictive, and prescriptive analytics.

Typical architecture



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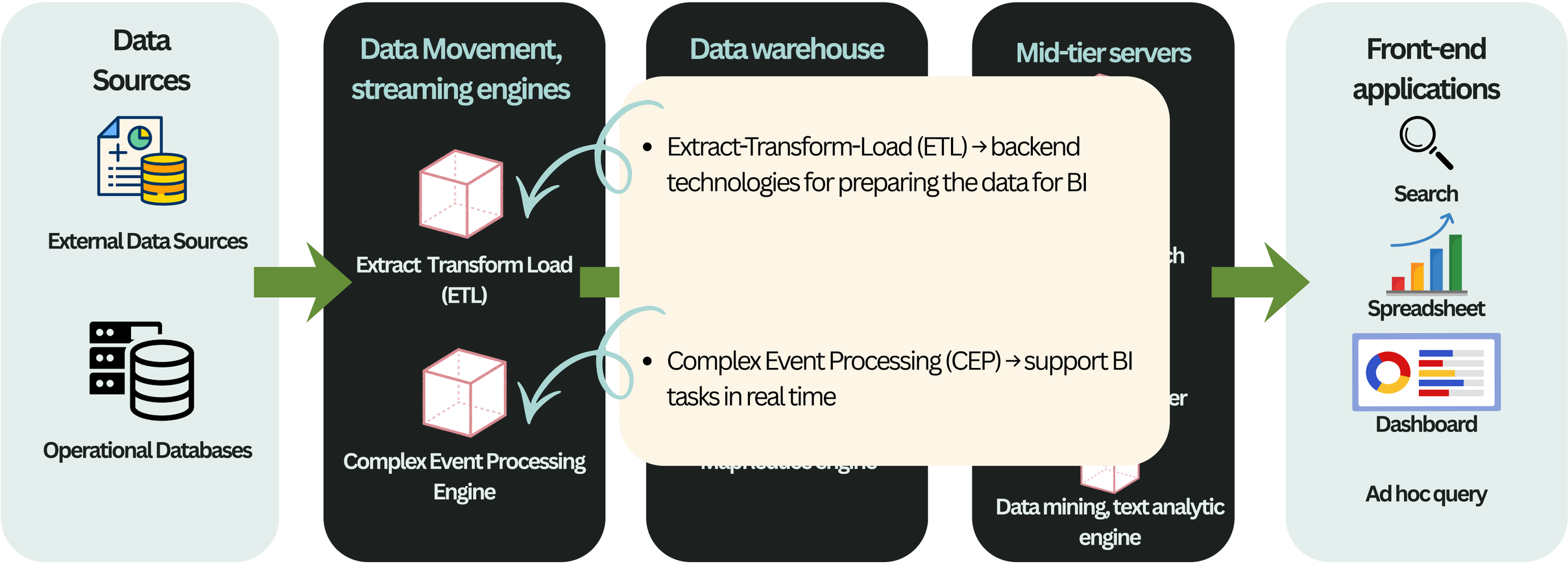
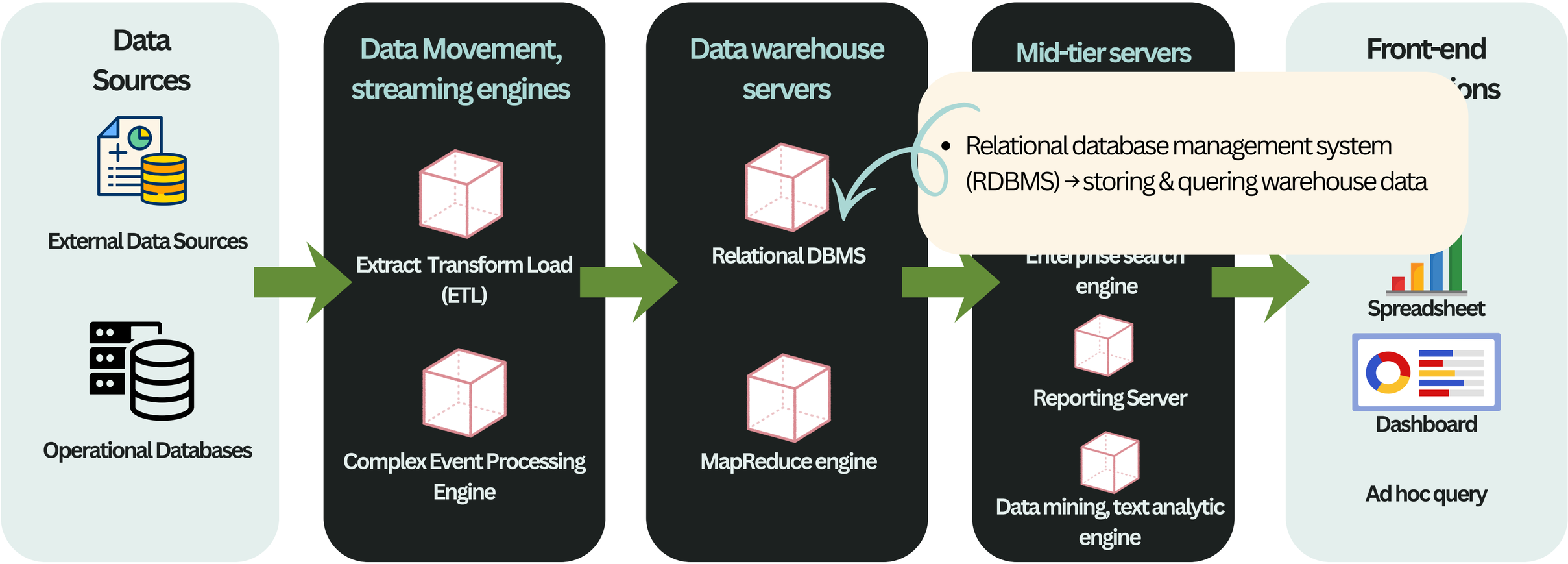


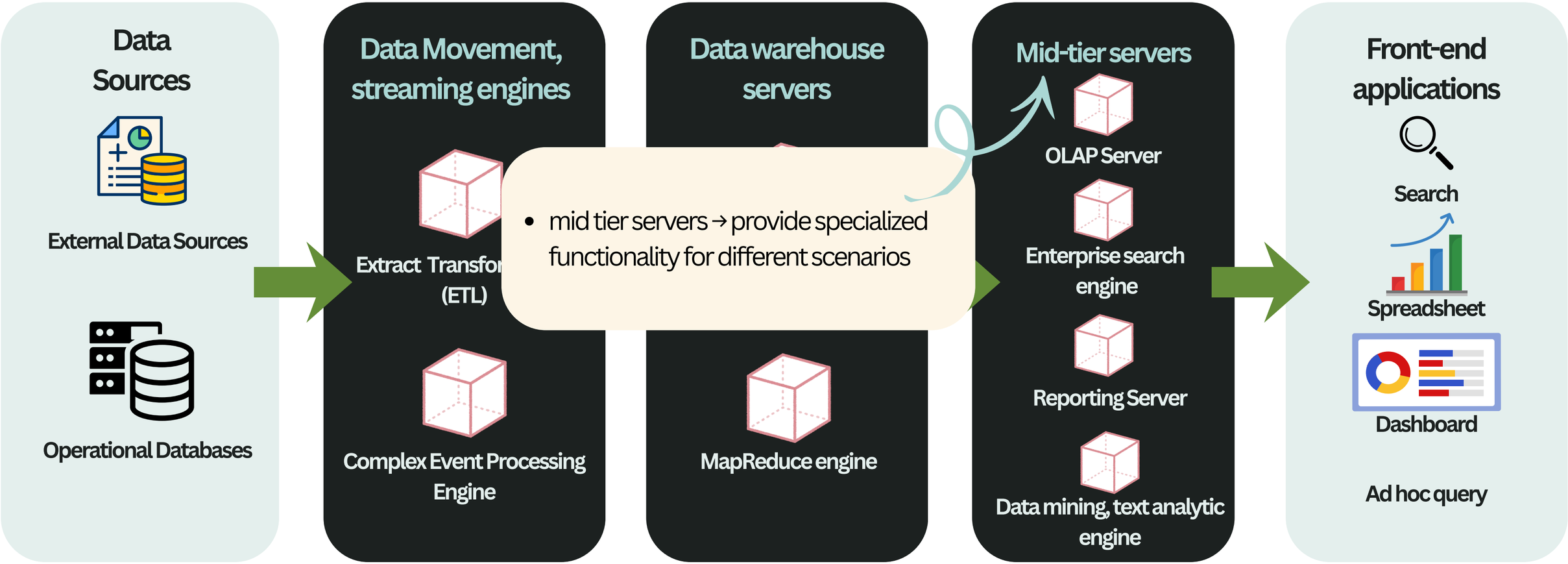
- Extract-Transform-Load (ETL) → backend technologies for preparing the data for BI
- Complex Event Processing (CEP) → support BI tasks in real time

Typical architecture



- Relational database management system (RDBMS) → storing & quering warehouse data

Typical architecture



- mid tier servers → provide specialized functionality for different scenarios

Different fields

(Big) Data Analytics

- BI&A technologies grounded in data mining & statistical analysis
- based on technologies of relational DBMS, data warehousing, ETL, OLAP & BPM
- Statistical machine learning based on mathematical models & powerful algorithms
- Hadoop & MapReduce have been revolutionary for large-scale, parallel data access

Text Analytics

- automatically extract specific kinds of structured information from documents
- Multimedia information retrieval & mobile information retrieval

Web Analytics

- HTTP/HTML-based hyperlinked web sites & associated web search enginces & directory systems for locating web content → web page updating/ranking
- development of cloud computing platforms & services → applications, system software, hardware delivered as services over the internet
- enables understanding of how visitors to a company's eb site interact wiht the pages

Network Analysis

- grounded in bibliometric analysis, citation networks
- link mining & community detection

Mobile Analysis

- ability to collect fine-grained, location-specific, context-aware, highly personalized content through smart devices
- opportunities for enabling BI applications for knowledge workers on mobile devices

Scopes of Analytics

Descriptive

What happened? What is happening?

- Business reporting
- Dashboards
- Scorecards
- Data Warehousing

Well defined business problems & opportunities

Predictive

What will happen? Why?

- Data Mining
- Text Mining
- Web/Media Mining
- Machine Learning

Accurate projections of future events and outcomes

Prescriptive

What should I do? Why?

- Optimization
- Simulation
- Decision Modeling
- Network Science

Best possible business decisions and actions

Business Intelligence

Advanced Analytics

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The purpose of BI&A

Big Data Explosion
Manage vast, diverse data volumes

Global Competition
Drive precise, rapid strategic decisions

Affordable Technology
Leverage cloud & AI for advanced analytics

Cultural Shift
Transition to evidence-driven insights

Regulatory Pressure
Ensure robust data governance & compliance

The purpose of BI&A

- leverage opportunities presented by the abundant data
- domain-specific analytics needed
- develop the appropriate analytical techniques to derive the intended impact

- to describe, predict, and optimise business performance
- insight they need to make better and faster decisions, which would improve their competitive posture in the marketplace
- drive business decisions & deliver new functionality (personalized offers, improved customer service etc.)

Use Cases

web analytics tools like Google Analytics can provide a trail of the user's online activities and **reveal the user's browsing & purchasing patterns**

Significant amount of unstructured content collected by an organization is textual format → **need for text analytics**

mobile computing → channel for reaching many users and increases the productivity & efficiency of an organization's workforce

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General Application

- Business-centric practices and methodologies
- **Manufacturing →** order shipment & customer support
- **Retail →** product pricing & user profiling to target grocery coupons during checkout, customer relationship management
- **Financial Services →** Risk mitigation, claims analysis & fraud detection
- **Transportation →** Fleet management
- **telecommunication →** Identifying reasons for customer churn
- **Utilities →** Power usage analysis
- **Healthcare →** Outcome anaylsis

Business & Marketing

- Website design, product placement optimization, customer transaction analysis, market structure analysis, product recommendation systems → **‘listen’ to the voice of the market**

- **customer relationship management** where sophisticated models are developed to identify new customers & up-sell/cross-sell opportunities

- businesses are trying to stay on top of what people are saying about their product/services and brands. → **social media analytics, text & sentiment analysis**

Science & Technology

- managing, analyzing, visualizing, & extracting useful information from large, diverse, distributed data sets

- accelerate progress of scientific discovery
- open source data processing & analytics tools

- Hypothesis testing
- Knowledge discovery

Health Care & Wellbeing

- electronic health records → significant clinical knowledge & deeper understanding of patient disease patterns

- research association in medical diagnosis & temporal relations between events

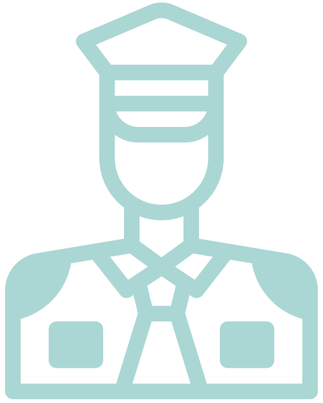
- decision support & patient empowerment
- patient community analysis
- health care decision support
- outcome analysis

Security & Public Safety

Development of advanced information technologies, systems, algorithms, & databases for security-related applications

- criminal association rule mining & clustering
- Crime analysis
- Terrorism informatics

- cyber attack analysis & attribution



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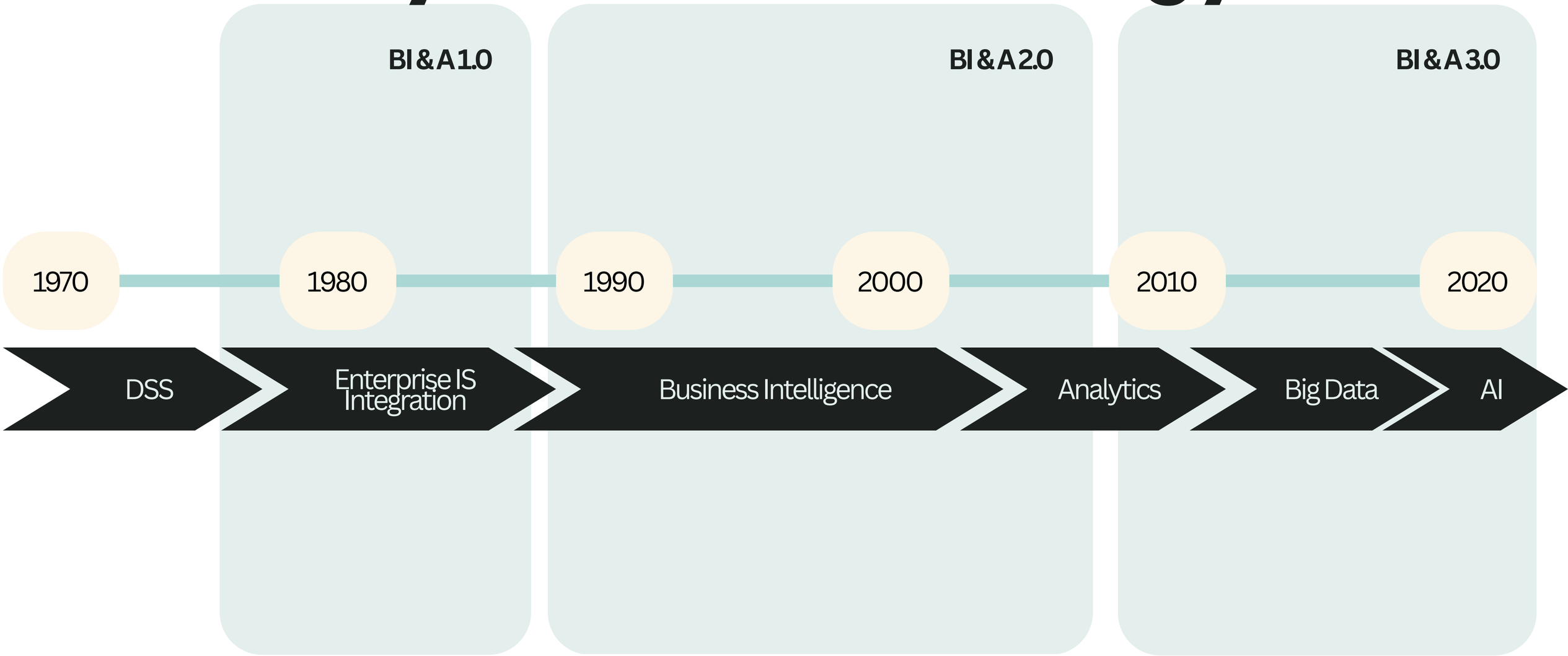
The storyline & Terminology



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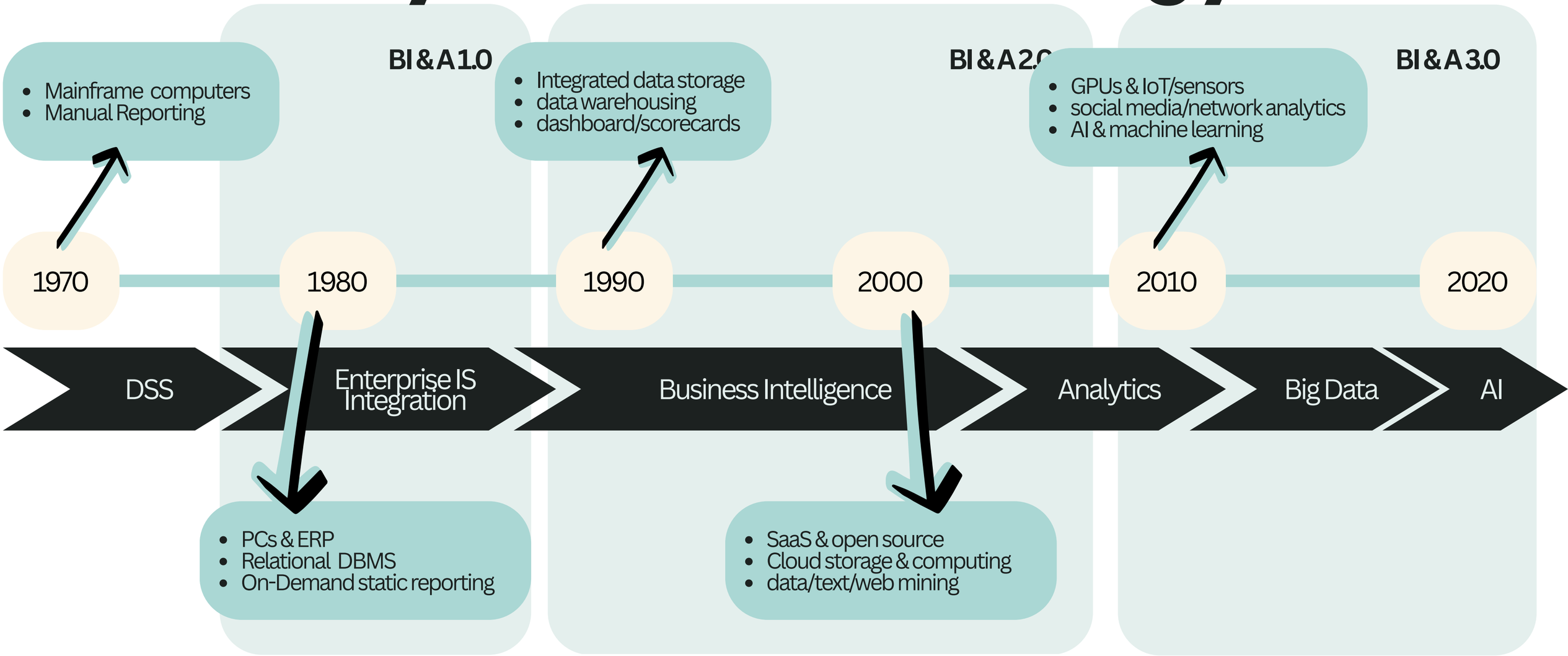
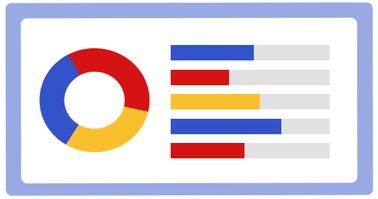
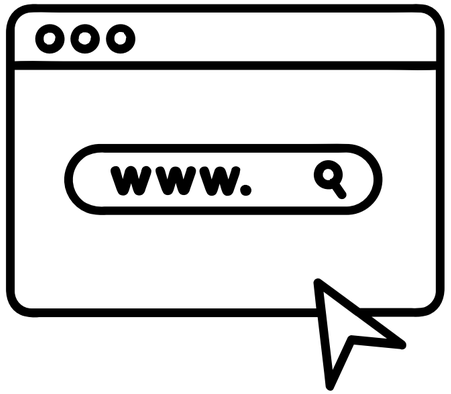


BI&A 1.0



- roots in **database management** & usage of **structured content**
- tools for extraction, transformation, and load (ETL) are essential for converting and integrating **company-specific data**
- Database query, online analytical processing (OLAP) and reporting tools based on simple graphics
- **Business performance management (BPM)** using scorecards & dashboards help analyze & visualize performance metrics → business recorting became business practice

BI&A 2.0

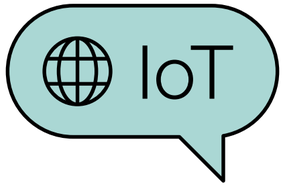


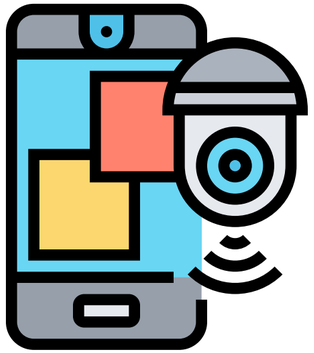
- Web-based, unstructured content
- Internet & web offer unique data collection & analytical research

- businesses present their businesses online & interact directly with customers
- applications can efficiently gather a large volume of feedback & opinios from diverse customer population → **opinion mining, information retrieval & extraction**
- shift from intuition-driven decision-making to evidence-based decision-making

- most large to medium size businesses adopted **data warehousing (DW)** as platform for decision-making
- dashboards and scorecards get data from the DW → not hindering the efficiency of the ERP systems

BI&A 3.0





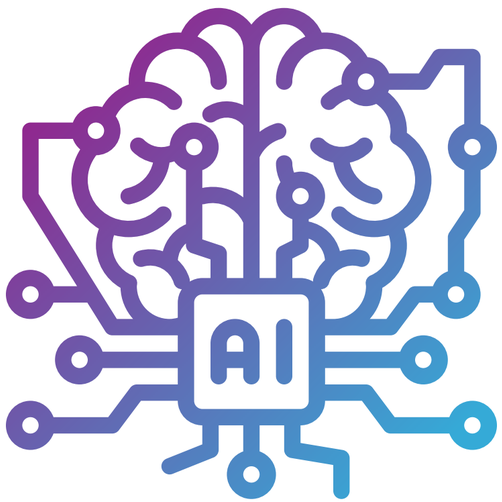
- Mobiles, IoT & sensors are opening up exciting new steams of **innovative applications**

- Automated data collections systems increased the quantity and quality of organisational data
- declining cost of acquiring and storing very large amounts of data
- New massively parallel data architecturs & analytic tools

- Mobile interface, visualization, HCI (human-computer interaction) design are also promising areas

- Competitive pressure has led to the increased need for near real time BI → **competition is now global**
- Increasing demand to deliver interactive BI experiences on mobile devices for today's knowledge workers

What about now?



- **AI-powered BI tools** are increasingly embedded in every part of a business
  - operate more intelligently,
  - predict trends with higher accuracy,
  - make data-driven decisions in real-time

- In the **future**:
  - more transformative changes
  - advances in natural language processing, computer vision, and reinforcement learning
  - **understand context, make autonomous decisions, and continuously learn and improve**

Thank you very much!

QUESTIONS?

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